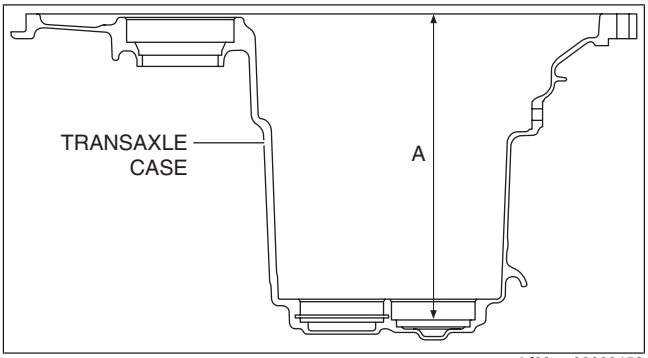
SHIM ADJUSTMENT

id051500285000

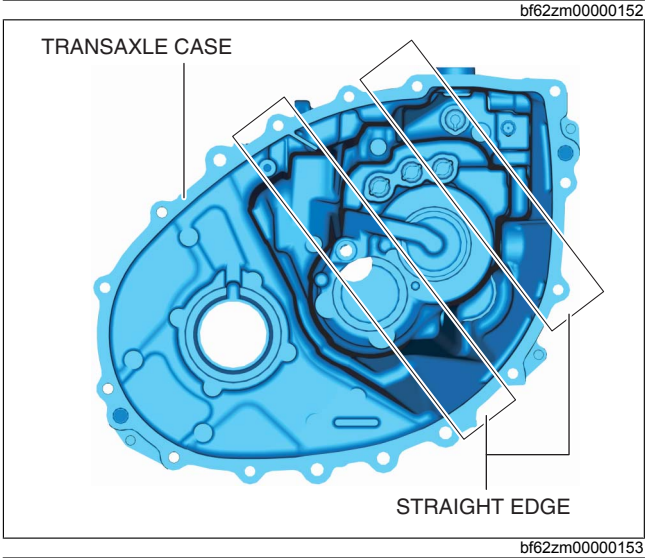
Primary Shaft Shim Adjustment

1. Measure the thickness of the straight edge which is used for measurements.
2. Measure distance A of the transaxle case using the following procedure.



bf62zm00000152

(1) Place two straight edges on the transaxle case as shown in the figure.

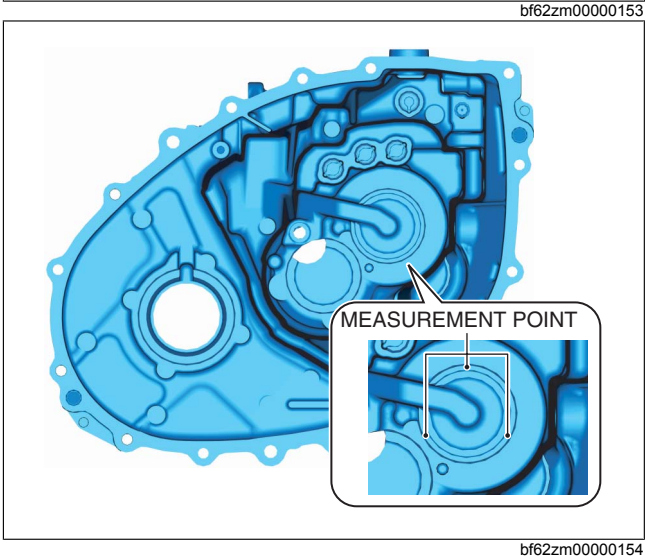


bf62zm00000153

(2) Set up 3 measurement points of your choosing on the shim assembly area of the clutch housing as shown in the figure.
(3) Using a commercially available depth gauge, measure the distance to each measurement point from the straight edge.

Caution

- **When measuring the distance to the measurement point from the straight edge, do not allow the depth gauge head to contact the shim assembly area.**



bf62zm00000154

(4) Calculate the average value measured in Step 3 using the following formula.
   - (Value of first measurement + 2 value of second measurement + 3 value of third measurement) / 3 = Average measured values.

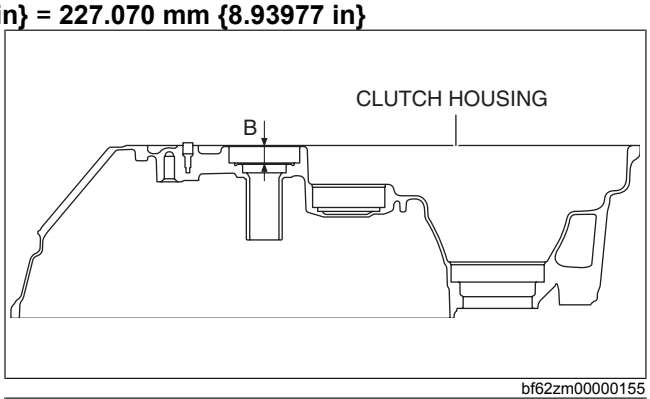
   Calculation example
   - First measured value = **232.069 mm {9.13658 in}**
   - Second measured value = **232.071 mm {9.13666 in}**
   - Third measured value = **232.070 mm {9.13662 in}**
   - (**232.069 mm {9.13658 in}** + **232.071 mm {9.13666 in}** + **232.070 mm {9.13662 in}**) / 3 = **232.070 mm {9.13662 in}**

(5) Calculate distance A of the transaxle case using the following formula.
   - Average of measured values - straight edge thickness = Distance A

   Calculation example
   - Straight edge thickness = **5.000 mm {0.1969 in}**
   - Measured average value = **232.070 mm {9.13662 in}**

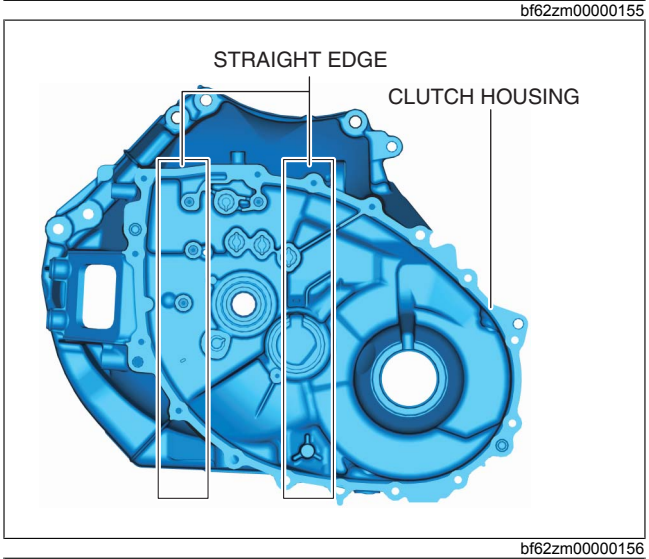
- **232.070 mm {9.13662 in} - 5.000 mm {0.1969 in} = 227.070 mm {8.93977 in}**

3. Measure distance B of the transaxle case using the following procedure.



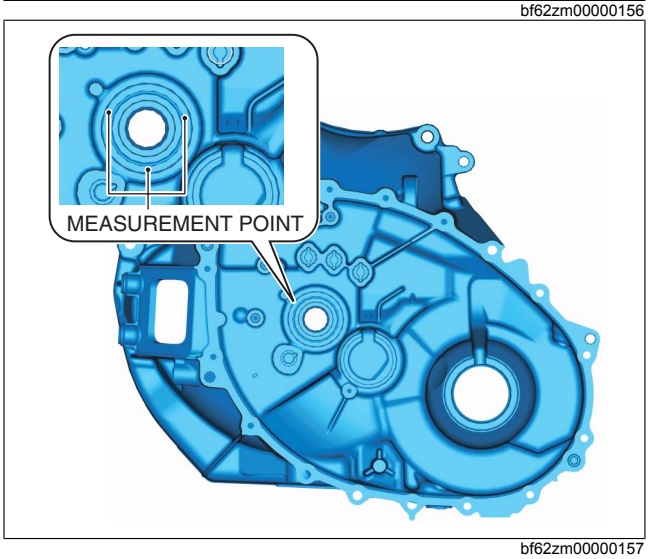
bf62zm00000155

(1) Set two straight edges on the clutch housing as shown in the figure.



bf62zm00000156

(2) Set up 3 measurement points of your choosing on the ball bearing assembly area of the clutch housing as shown in the figure.
(3) Using a commercially available depth gauge, measure the distance to each measurement point from the straight edge.



bf62zm00000157

Caution
- **When measuring the distance to the measurement points from the straight edge, do not allow the depth gauge head to contact the ball bearing assembly area.**

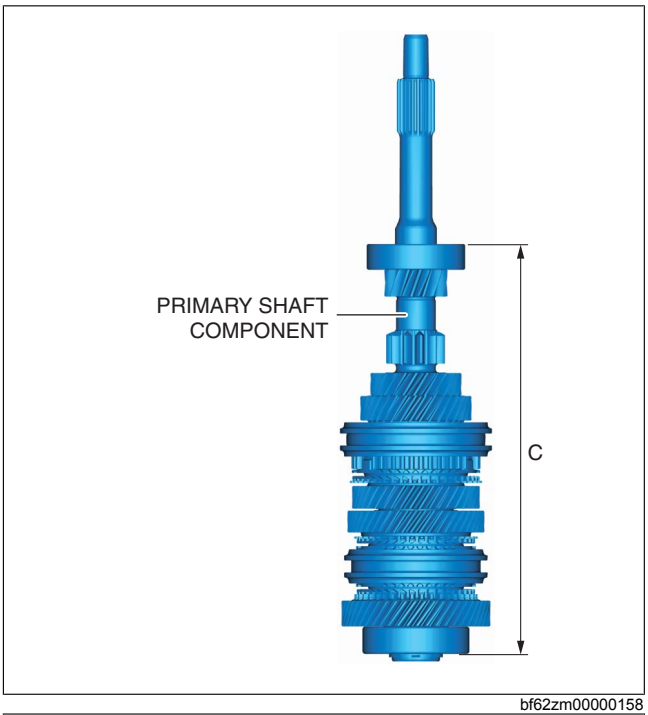
(4) Calculate the average value measured in Step 3 using the following formula.
- (Value of first measurement + 2 value of second measurement + 3 value of third measurement) / 3 = Average measured values.

Calculation example
- First measured value = **19.001 mm {0.74807 in}**
- Second measured value = **18.999 mm {0.74799 in}**
- Third measured value = **19.000 mm {0.74803 in}**
- (**19.001 mm {0.74807 in}** + **18.999 mm {0.74799 in}** + **19.000 mm {0.74803 in}**) / 3 = **19.000 mm {0.74803 in}**

(5) Calculate distance B of the clutch housing using the following formula.
- Average of measured values - straight edge thickness = Distance B

Calculation example
- Straight edge thickness = **5.000 mm {0.1969 in}**
- Measured average value = **19.000 mm {0.74803 in}**
- **19.000 mm {0.74803 in}** - **5.000 mm {0.1969 in}** = **14.000 mm {0.55118 in}**

4. Measure distance C of the primary shaft component using the following procedure.

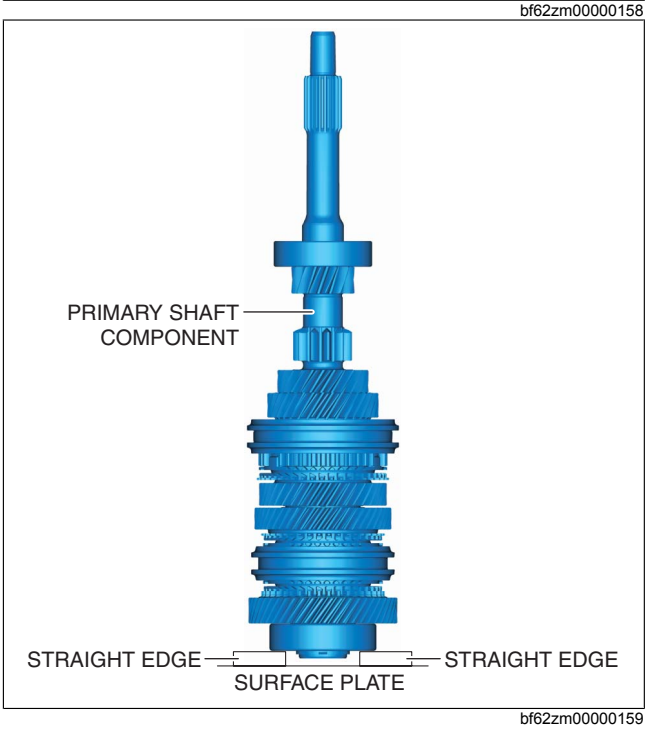


bf62zm00000158

(1) Place two straight edges on the surface plate as shown in the figure and set the primary shaft component on top of them.

Caution

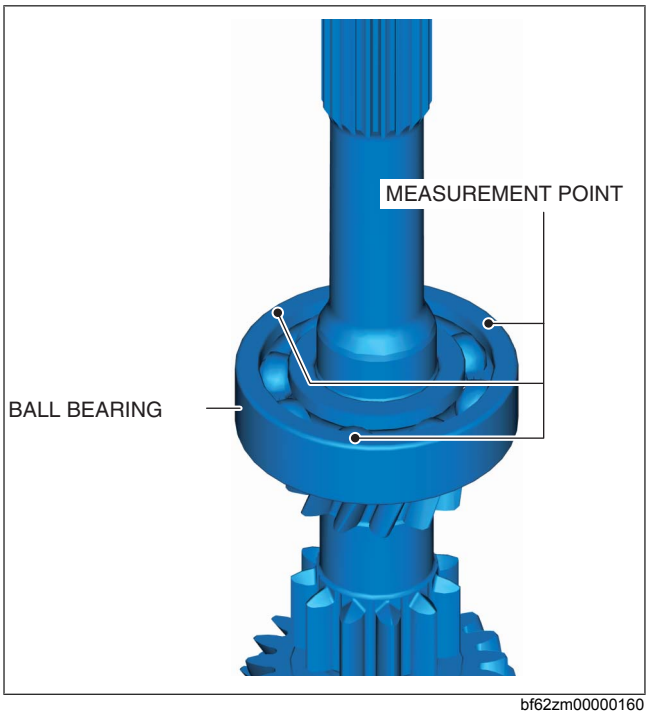
- **To measure distance C correctly, set the primary shaft component so that the straight edge does not contact the ball bearing inner race.**



bf62zm00000159

(2) Set up 3 measurement points of your choosing on the ball bearing as shown in the figure.
(3) Using a commercially available height gauge, measure the distance to each measurement point from the straight edge.
(4) Calculate the average of the value measured in Step 3 using the following formula, and use the average of the measured value for distance C of the primary shaft component.
- (Value of first measurement + value of second measurement + value of third measurement) / 3 = Distance C

Calculation example
- First measured value = **239.706 mm {9.43725 in}**
- Second measured value = **239.705 mm {9.43721 in}**
- Third measured value = **239.707 mm {9.43729 in}**
- (**239.706 mm {9.43725 in}** + **239.705 mm {9.43721 in}** + **239.707 mm {9.43729 in}**) / 3 = **239.706 mm {9.43725 in}**



bf62zm00000160

5. Select the primary shaft shim using the following procedure.
(1) Calculate the clearance (clearance D) between the transaxle case and the primary shaft component using the following formula.
- Distance A + Distance B - Distance C = Clearance D

Calculation example
- Distance A = **227.070 mm {8.93977 in}**
- Distance B = **14.000 mm {0.55118 in}**
- Distance C = **239.706 mm {9.43725 in}**
- **227.070 mm {8.93977 in}** + **14.000 mm {0.55118 in}** - **239.706 mm {9.43725 in}** = **1.364 mm {0.05370 in}**

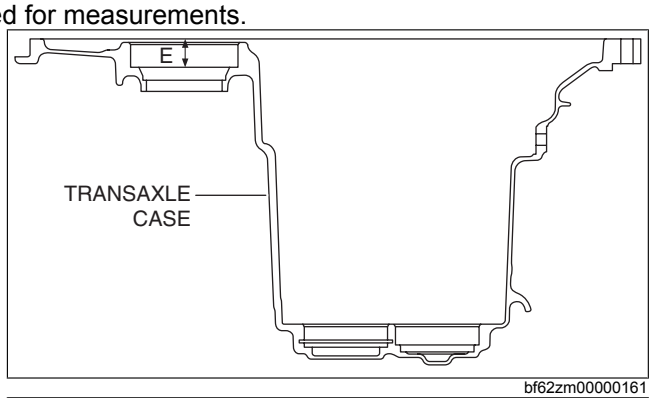
(2) Compare the calculated value for clearance D with the following table and select a shim of the appropriate thickness.

Clearance D measured value		Appropriate shim thickness (mm {in})
Calculated shim thickness of more (mm {in})	Less than calculated shim thickness (mm {in})	
1.570 {0.06181}	1.610 {0.06339}	1.550 {0.06102}
1.520 {0.05984}	1.570 {0.06181}	1.500 {0.05906}
1.470 {0.05787}	1.520 {0.05984}	1.450 {0.05709}
1.420 {0.05591}	1.470 {0.05787}	1.400 {0.05512}
1.370 {0.05394}	1.420 {0.05591}	1.350 {0.05315}
1.320 {0.05197}	1.370 {0.05394}	1.300 {0.05118}
1.270 {0.05000}	1.320 {0.05197}	1.250 {0.04921}
1.220 {0.04803}	1.270 {0.05000}	1.200 {0.04724}
1.170 {0.04606}	1.220 {0.04803}	1.150 {0.04528}
1.120 {0.04409}	1.170 {0.04606}	1.100 {0.04331}
1.070 {0.04213}	1.120 {0.04409}	1.050 {0.04134}
1.020 {0.04016}	1.070 {0.04213}	1.000 {0.03937}
0.970 {0.03819}	1.020 {0.04016}	0.950 {0.03740}
0.920 {0.03622}	0.970 {0.03819}	0.900 {0.03543}
0.870 {0.03425}	0.920 {0.03622}	0.850 {0.03346}
0.820 {0.03228}	0.870 {0.03425}	0.800 {0.03150}
0.770 {0.03031}	0.820 {0.03228}	0.750 {0.02953}
0.720 {0.02835}	0.770 {0.03031}	0.700 {0.02756}
0.670 {0.02638}	0.720 {0.02835}	0.650 {0.02559}
0.620 {0.02441}	0.670 {0.02638}	0.600 {0.02362}

Selection example
- For clearance D **1.364 mm {0.05370 in}**, select a shim with a thickness of **1.300 mm {0.05118 in}** because the calculated shim thickness value is **1.320 mm {0.05197 in}** or more and less than **1.370 mm {0.05394 in}**.

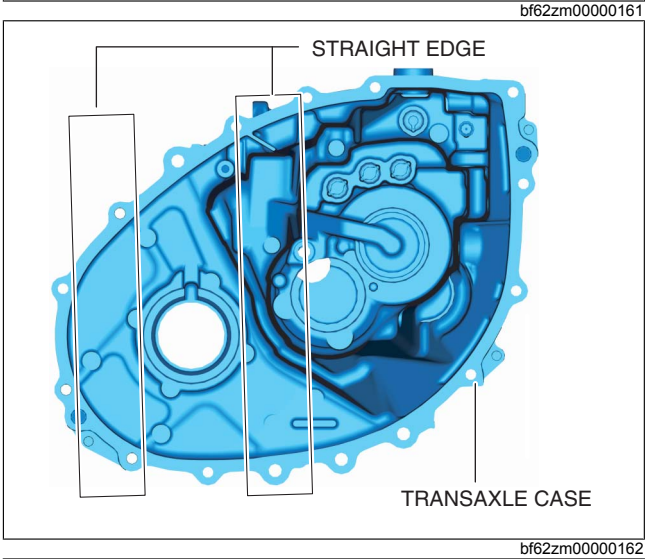
Differential Shim Adjustment

1. Measure the thickness of the straight edge which is used for measurements.
2. Measure distance E of the transaxle case using the following procedure.



bf62zm00000161

(1) Place two straight edges on the transaxle case as shown in the figure.

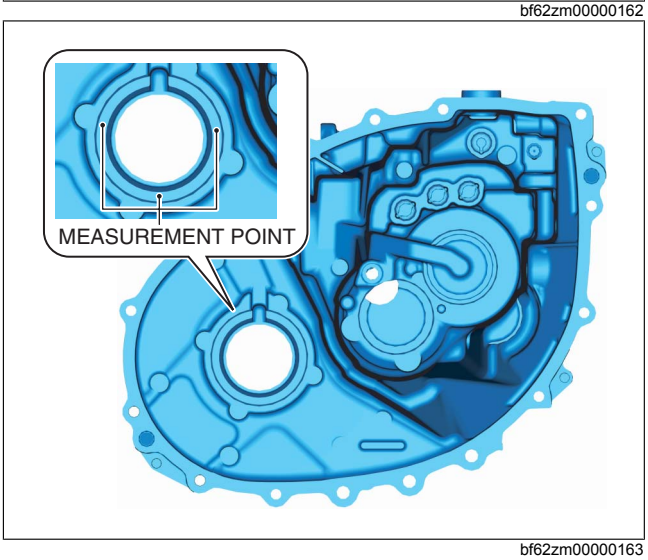


bf62zm00000162

(2) Set up 3 measurement points of your choosing on the shim assembly area of the transaxle case as shown in the figure.
(3) Using a commercially available depth gauge, measure the distance to each measurement point from the straight edge.

Caution

- **When measuring the distance to the measurement point from the straight edge, do not allow the depth gauge head to contact the shim assembly area.**



bf62zm00000163

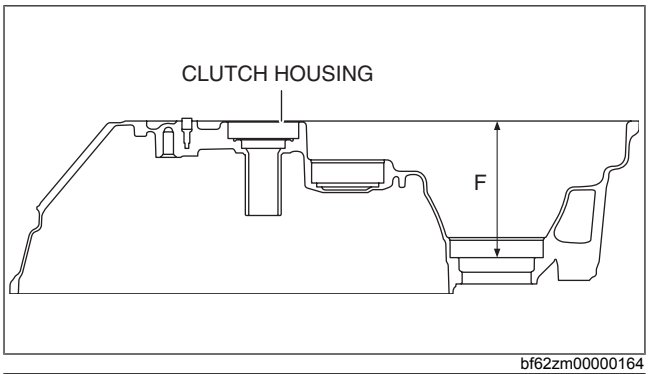
(4) Calculate the average value measured in Step 3 using the following formula.
   - (Value of first measurement + 2 value of second measurement + 3 value of third measurement) / 3 = Average measured values.

   Calculation example
   - First measured value = **25.901 mm {1.0197 in}**
   - Second measured value = **25.899 mm {1.0196 in}**
   - Third measured value = **25.900 mm {1.0197 in}**
   - (**25.901 mm {1.0197 in}** + **25.899 mm {1.0196 in}** + **25.900 mm {1.0197 in}**) / 3 = **25.900 mm {1.0197 in}**

(5) Calculate distance E of the transaxle case using the following formula.
   - Average of measured values - straight edge thickness = Distance E

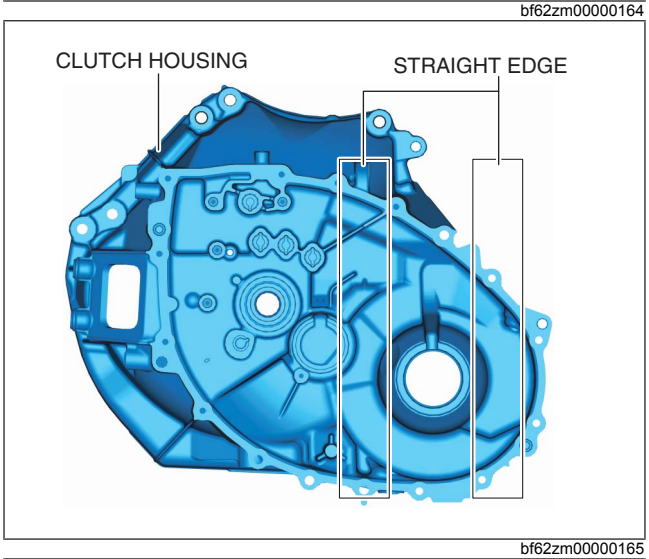
   Calculation example
   - Straight edge thickness = **5.000 mm {0.1969 in}**
   - Measured average value = **25.900 mm {1.0197 in}**
   - **25.900 mm {1.0197 in}** - **5.000 mm {0.1969 in}** = **20.900 mm {0.82284 in}**

3. Measure distance F of the transaxle case using the following procedure.



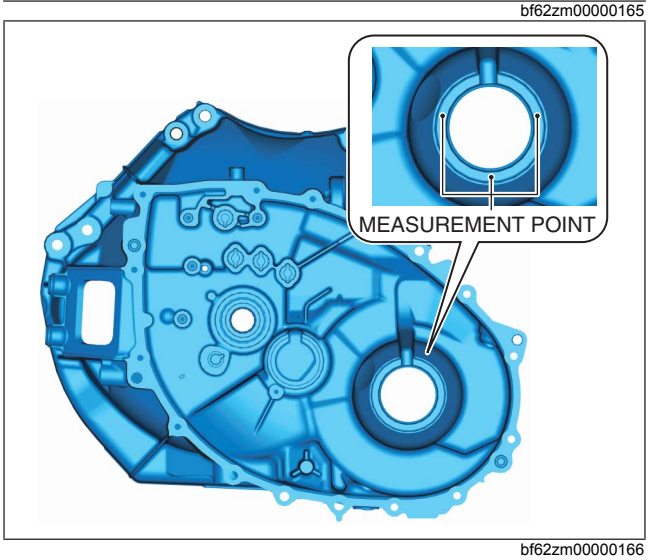
bf62zm00000164

(1) Set two straight edges on the clutch housing as shown in the figure.



bf62zm00000165

(2) Set up 3 measurement points of your choosing on the ball bearing assembly area of the clutch housing as shown in the figure.
(3) Using a commercially available depth gauge, measure the distance to each measurement point from the straight edge.



bf62zm00000166

Caution
- **When measuring the distance to the measurement point from the straight edge, do not allow the depth gauge head to contact the bearing outer race assembly area.**

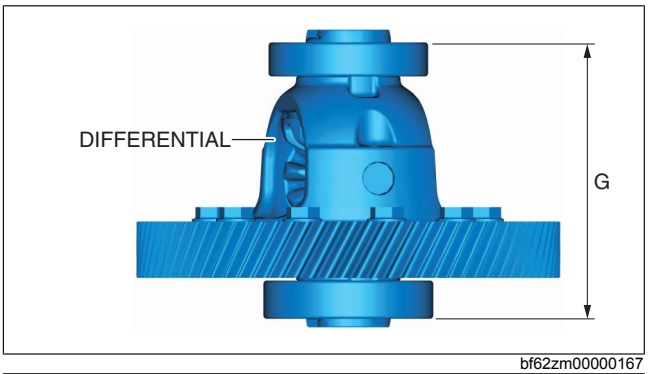
(4) Calculate the average value measured in Step 3 using the following formula.
- (Value of first measurement + 2 value of second measurement + 3 value of third measurement) / 3 = Average measured values.

Calculation example
- First measured value = **106.099 mm {4.17713 in}**
- Second measured value = **106.100 mm {4.17717 in}**
- Third measured value = **106.101 mm {4.17721 in}**
- (**106.099 mm {4.17713 in}** + **106.100 mm {4.17717 in}** + **106.101 mm {4.17721 in}**) / 3 =**106.100 mm {4.17717 in}**

(5) Calculate distance F of the clutch housing using the following formula.
- Average of measured values - straight edge thickness = Distance F

Calculation example
- Straight edge thickness = **5.000 mm {0.1969 in}**
- Measured average value = **106.100 mm {4.17717 in}**
- **106.100 mm {4.17717 in}** - **5.000 mm {0.1969 in}** = **101.100 mm {3.98032 in}**

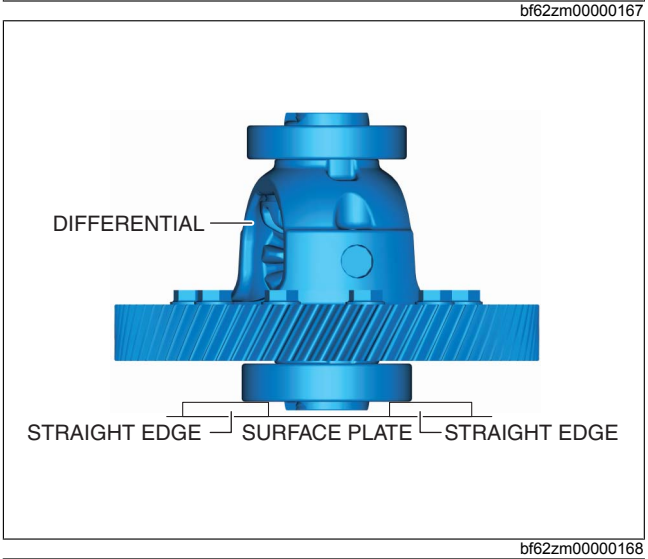
4. Measure distance G of the differential using the following procedure.



bf62zm00000167

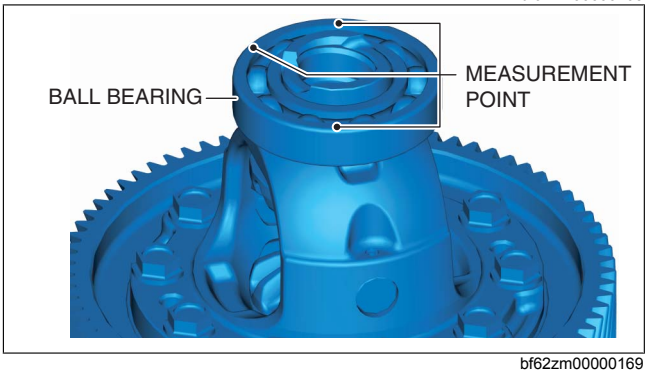
(1) Place two straight edges on the surface plate as shown in the figure and set the differential on top of them.

Caution
- **To measure distance G correctly, set the differential so that the straight edge does not contact the ball bearing inner race.**



bf62zm00000168

(2) Set up 3 measurement points of your choosing on the ball bearing as shown in the figure.
(3) Using a commercially available height gauge, measure the distance to each measurement point from the straight edge.
(4) Calculate the average of the value measured in Step 4 using the following formula, and use the average of the measured value for distance G of the differential.
- (Value of first measurement + 2 value of second measurement + 3 value of third measurement) / 3 = Distance G



bf62zm00000169

Calculation example
- First measured value = **120.898 mm {4.75977 in}**
- Second measured value = **120.899 mm {4.75981 in}**
- Third measured value = **120.897 mm {4.75973 in}**
- (**120.898 mm {4.75977 in}** + **120.899 mm {4.75981 in}** + **120.897 mm {4.75973 in}**) / 3 = **120.898 mm {4.75977 in}**

5. Select the differential shim using the following procedure.
(1) Calculate the clearance (clearance H) between the transaxle case and the primary shaft component using the following formula.
- Distance E + Distance F - Distance G = Clearance H

Calculation example
- Distance E = **20.900 mm {0.82284 in}**
- Distance F = **101.100 mm {3.98032 in}**
- Distance G = **120.898 mm {4.75977 in}**
- **20.900 mm {0.82284 in}** + **101.100 mm {3.98032 in}** - **120.898 mm {4.75977 in}** = **1.102 mm {0.04339 in}**

(2) Compare the calculated value for clearance H with the following table and select a shim of the appropriate thickness.

Clearance H measured value		Appropriate shim thickness (mm {in})
Calculated shim thickness of more (mm {in})	Less than calculated shim thickness (mm {in})	
1.620 {0.06378}	1.670 {0.06575}	1.600 {0.06299}

Clearance H measured value		Appropriate shim thickness (mm {in})
Calculated shim thickness of more (mm {in})	Less than calculated shim thickness (mm {in})	
1.570 {0.06181}	1.620 {0.06378}	1.550 {0.06102}
1.520 {0.05984}	1.570 {0.06181}	1.500 {0.05906}
1.470 {0.05787}	1.520 {0.05984}	1.450 {0.05709}
1.420 {0.05591}	1.470 {0.05787}	1.400 {0.05512}
1.370 {0.05394}	1.420 {0.05591}	1.350 {0.05315}
1.320 {0.05197}	1.370 {0.05394}	1.300 {0.05118}
1.270 {0.05000}	1.320 {0.05197}	1.250 {0.04921}
1.220 {0.04803}	1.270 {0.05000}	1.200 {0.04724}
1.170 {0.04606}	1.220 {0.04803}	1.150 {0.04528}
1.120 {0.04409}	1.170 {0.04606}	1.100 {0.04331}
1.070 {0.04213}	1.120 {0.04409}	1.050 {0.04134}
1.020 {0.04016}	1.070 {0.04213}	1.000 {0.03937}
0.970 {0.03819}	1.020 {0.04016}	0.950 {0.03740}
0.920 {0.03622}	0.970 {0.03819}	0.900 {0.03543}
0.870 {0.03425}	0.920 {0.03622}	0.850 {0.03346}
0.835 {0.03287}	0.870 {0.03425}	0.800 {0.03150}

Selection example

- For clearance H **1.102 mm {0.04339 in}**, select a shim with a thickness of **1.050 mm {0.04134 in}** because the calculated shim thickness value is **1.070 mm {0.04213 in}** or more and less than **1.120 mm {0.04409 in}**.